Project: Monte Carlo simulations at constant N, P, T

Objective: To create a Monte Carlo (MC) code for (N,P,T) simulations of Lennard-Jones (LJ) systems.

Outline

1. Write a MC code for a (N,P,T) simulation of a Lennard-Jones (LJ) system. Use N=500 atoms. Remember that the LJ potential is of the form: $U_{LJ}(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$.
2. Run a series a series of MC simulations at $T^* = 2.0$ and pressures from $P^* = 0.15$ to $P^* = 15$.
3. Compute the density ρ^*, the interaction energy U^*, the pressure P^* and the radial distribution function $g(r)$ of the system.
4. Plot the quantities U^* and P^* (with error bars) as a function of ρ^*. Compare the present results with those obtained from MC simulations at constant N, V and T.
5. Plot the $g(r)$ as a function of pressure.

Lennard-Jones parameters

atom	mass(uma)	ϵ/k_B (K)	σ ($\mathring{A}$)
Ar	39.948	119.8	3.405

Questions

1. **Summarize the Metropolis algorithm for (N,P,T) simulations. Describe the main hypothesis involved in its derivation.**
2. Define the reduced units used in your simulations. Translate from reduced units to real units the values of T^*, ρ^*, U^* and P^*.
3. Describe how did you fix the parameters involved in the trial changes of the volume.
4. Derive the expressions for the tail corrections to the energy and pressure ($U_{LJ}^{*,tail}$, $P_{LJ}^{*,tail}$).